

popquiz momentum2 - ⚡

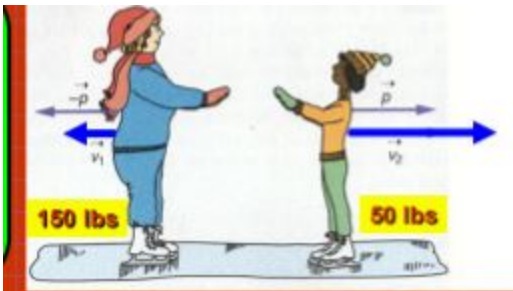
⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:43pm

Quiz Instructions



Question 1 20 pts



Amy (150 lbs) and Gwen (50 lbs) are standing on slippery ice and push off each other. If Amy slides at 6 ft/s, what speed does Gwen have?

hint: no conversion necessary.



18ft/s



9ft/s



6ft/s



12ft/s



Question 2 20 pts

A small car and a large truck collide head-on and stick together. Which one has the larger momentum change?

Hint: remember mowgli and baloo discussion. Conservation of momentum is a consequence of Newton's third law.

- ☐ the car
- ☐ They both have the same momentum change
- ☐ Not enough information. Missing final speed.
- ☐ the truck



Question 3 20 pts

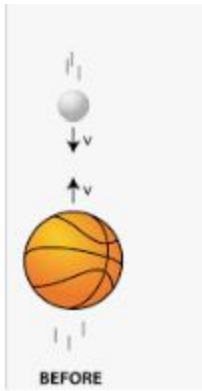
A small beanbag and a bouncy rubber ball are dropped from the same height above the floor. They both have the same mass. Which one will impart the greater impulse to the floor when it hits?

Hint: impulse is the change in momentum . see my slide. the bouncing ball and the crashing teddy bear.

- ☐ both the same
- ☐ the beanbag
- ☐ the rubber ball
- ☐ whaaaat ?



Question 4 20 pts



We discussed this demo in class. The basket ball is very very massive relative to the ping pong ball (inertia large $\rightarrow$ tiny change in velocity). The collision is elastic so the relative velocity of the balls before and after stays the same (so before relative speed is $2V$. head on collision).

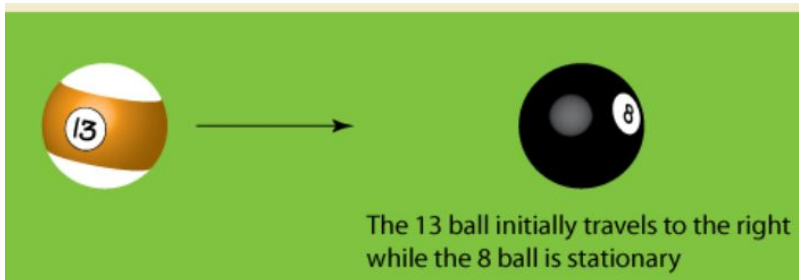
before: basketball has a velocity $V @$ up and pinpong has the velocity $V @$ down.

After?

- ☐ basketball $V @$ up and pinpong stops
- ☐ basketball $V @$ up and pinpong $3V @$ up
- ☐ basket stops and pingpong $2V @$ up
- ☐ basketball $V @$ up and pinpong $V @$ up



Question 5 20 pts



What happens next ? The collision is elastic. They have the same mass. They collide along a line. (1D)

unsure? https://www.walter-fendt.de/html5/phen/collision_en.htm

- ☐ I missed the class
- ☐ both bounce off each other

☐

13th bounces off and 8th stops

☐

13th stops and 8th moves @ right

Not saved

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